# *Local Income Taxes and Residential Sorting*
## *Evidence from Pennsylvania LEHD Origin–Destination Data*

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# Motivation

- Tiebout (1956): citizens sort across jurisdictions according to their demands for local public goods.
- Strong evidence of tax-driven sorting at the state level (**??**).
- Little causal evidence on local income tax sorting within metro areas
- Two data problems make this hard. Local tax jurisdictions are sub-county, so IRS/ACS migration data are too coarse. Credit systems and weak enforcement complicate the effective tax burden.

# This Paper

- Question: Do local income tax rate changes cause sorting?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract with workplace tract annually.
- Outcomes: `share_live_in` = (workers who live and work in city $i$) / (total workers in $i$), plus count outcomes (resident workers, total jobs).
- Exogeneous identification: Act 32 consolidated collection and made employer withholding mandatory.

# Institutional Setting (Pennsylvania)

- Universal Earned Income Tax. About 2,500 munis and 500 school districts each set their own rate. Typical 0.5% muni + 0.5% school district.
- Before Act 32, collection was fragmented and withholding was inconsistent.
- Act 32 (effective 2012) consolidated collection and required employers to withhold the higher of the home and work rates.

# How the PA EIT works (post-Act 32, excluding Philadelphia)

A worker pays the higher of the total home rate (muni resident plus school district) and the nonresident rate at the work location. The bill is split piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}} \end{cases}$$

- The work jurisdiction always collects its full nonresident rate.
- The home muni and school district then split whatever remains of the worker's bill. They get nothing if the work rate is higher.
- Which rate binds determines how the worker can respond. When the home rate binds, the worker can lower the bill by moving home. When the work rate binds, the worker can lower it only by changing jobs.

# Worked example

Worker 1 lives in city $B$ and works in city $A$. Home muni rate is 1.0%, home school district is 0.5%, so the total home rate is 1.5%. The work-location nonresident rate is 0.8%.

- Worker 1 pays the higher of the two rates, so 1.5%. Home collects $1.5 - 0.8 = 0.7\%$; work collects 0.8%.
- This is a home-binds case. To lower the bill, worker 1 would have to move out of $B$. Changing jobs would not help because the work rate is already lower.

# Philadelphia: a separate regime

- Philadelphia is exempt from Act 32. Its Wage Tax operates under the Sterling Act of 1932, administered by the City Department of Revenue.
- Philly residents pay the Philly resident rate regardless of where they work. There is no max rule and no offset.
- Nonresidents working in Philly pay the Philly nonresident rate, which goes entirely to Philly.
- In my analysis I keep Philly in the main sample because it is part of PA's labor market, but exclude it in robustness checks because its reporting and dynamics differ from Act-32 munis.

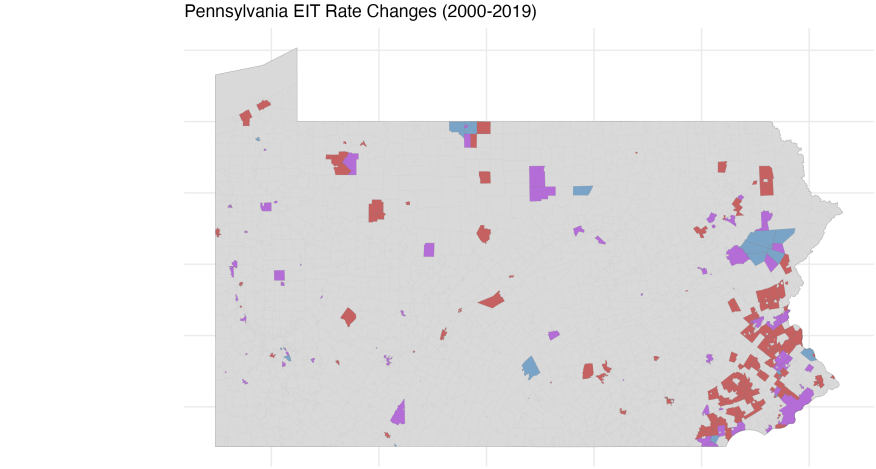
# Data Generating Process

1. Pull Census tract LODES OD flows (LEHD, 2002–2019).
2. Scrape PA rates from DCED MunStats: muni resident, muni nonresident, and school district EIT for each muni-year.
3. Scrape PA municipal Annual Financial Reports (DCED) for spending on police, fire, streets, sewer, parks, comm dev, debt, and state aid.
4. Crosswalk DCED munis to Census (99.6% match) and aggregate tract-pair flows to muni-pair-year.

# Summary Statistics (Pennsylvania, 2002–2019)

| Variable | Mean | Median | SD | Max |
|---|---|---|---|---|
| share_live_in | 0.103 | 0.088 | 0.076 | 0.657 |
| Muni resident EIT (%) | 0.532 | 0.500 | 0.175 | 4.331 |
| Muni nonres EIT (%) | 0.891 | 1.000 | 0.235 | 3.820 |
| SD EIT (%) | 0.617 | 0.500 | 0.289 | 2.050 |
| Total jobs | 2,438 | 565 | 14,502 | 629,698 |
| Resident workers | 450 | 50 | 8,001 | 391,047 |
| Muni × year | 35,743 (2,275 munis) | | | |

# Treatment Geography: PA Rate Changes




Red = increased; blue = decreased; purple = both; grey = unchanged.

# Three Channels of Adjustment

When muni $A$ raises its rate, residents of $A$ can:

1. Move residence to a lower-rate muni, keep job in $A$. Saves at least the home SD rate at $A$.
2. Move both residence and job to a lower-rate muni. Largest savings.
3. New entrants pick a home outside $A$ for a given job.

All three reduce `share_live_in` for $A$.

# Outcomes

Unit $i$ = PA municipality, $t$ = year, 2002–2019.

- $\texttt{share_live_in}_{it} = \dfrac{\text{live + work in } i}{\text{total workers in } i}$
- $\log(\text{Resident workers}_{it})$ = workers who live in muni $i$, any work location.
- $\log(\text{Total jobs}_{it})$ = workers employed in muni $i$, any home location.

# Q2: Do rate increases cause sorting?

## Two specs

$$\text{Total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

$$\text{Decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

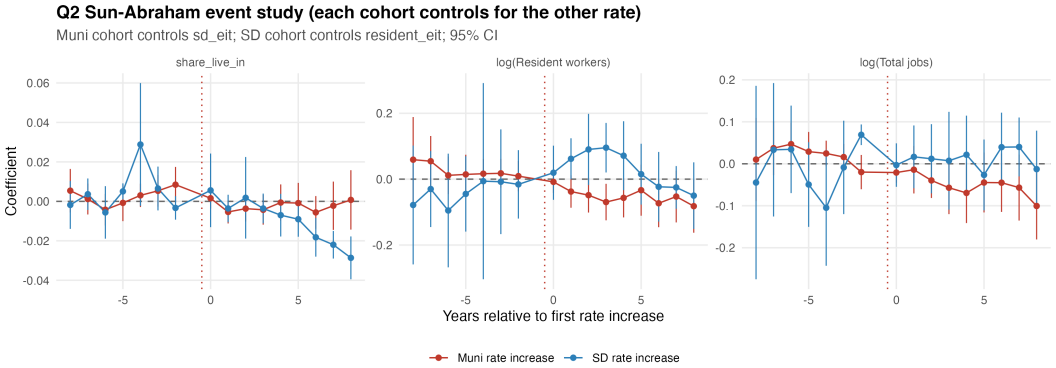
- Treated = munis that ever raised their muni or SD rate (within-city rate change drives identification).
- Control = munis whose rates stayed flat over the panel.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. $Y_{it}$ = `share_live_in` or log count outcomes.

# Q2 Answer: muni rate drives sorting; SD null

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| *Total stack* | | | |
| $\tau^{\text{home}}$ | $-0.010^{**}$ | $-0.137^{***}$ | $-0.096^{**}$ |
| | (0.004) | (0.043) | (0.038) |
| *Decomposition* | | | |
| $\tau^{\text{res,muni}}$ | $-0.012^{**}$ | $-0.201^{***}$ | $-0.128^{***}$ |
| | (0.006) | (0.041) | (0.038) |
| $\tau^{\text{SD}}$ | $-0.005$ | $+0.029$ | $-0.013$ |
| | (0.006) | (0.114) | (0.093) |
| $R^2$ | 0.909 | 0.975 | 0.976 |
| $N$ | 12,891 | 12,781 | 12,891 |

- Controls: log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- A 1 pp muni rate hike lowers share_live_in by 1.2 share-pp and resident workers by 20%.

# Q2 Event Study: muni vs SD decomposed



# Q2 DCDH: CONTINUOUS TREATMENT, STAGGERED DESIGN

de Chaisemartin & D'Haultfœuille (2024) continuous estimator handles staggered treatment + continuous dose without TWFE's negative-weighting problem.

$$Y_{it} = \alpha_i + \gamma_t + \text{ACR} \cdot \tau_{it}^{\text{res,muni}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

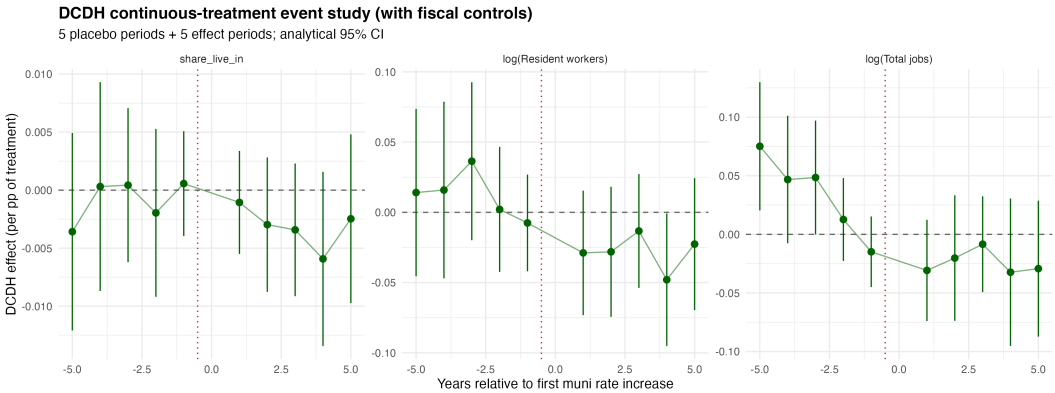
- Treated = munis ever raising muni rate.
- Control = munis whose rates stayed flat over the panel.
- Dose = muni resident rate (continuous).
- Effects up to $\ell = 5$; placebos up to $\ell = -5$.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. $Y_{it}$ = `share_live_in` or log count outcomes.

# Q2 DCDH Answer: average causal response per pp

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| ACR per pp | −0.009 | −0.078 | −0.067 |
| | (0.006) | (0.047) | (0.055) |
| $N$ (city-yrs) | 12,926 | 12,926 | 12,926 |
| Munis | 948 | 948 | 948 |

- **Note: DCDH not yet rerun on the block-aggregated panel** (computationally intense). Numbers above are from prior tract-centroid panel; expected to shift modestly toward the new TWFE muni point estimates ($-0.012^{**}$, $-0.201^{***}$, $-0.128^{***}$).
- DCDH addresses the staggered-DiD critique: no negative weights from already-treated comparison groups.

# Q2 DCDH Event Study



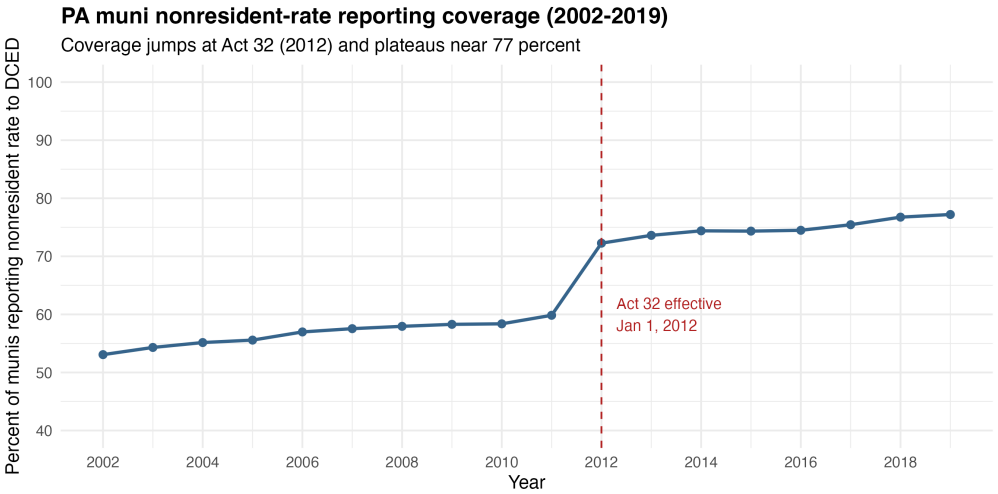
# Mechanism: Is it enforcement? (Act 32)

**Unit $i$ = PA municipality; $t$ = year.** Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

# Nonresident rate reporting coverage



# Do the rate increases we exploit actually bind?

**For self-pair (worker lives and works in muni $X$):** home stack $= \tau_X^{\text{res}} + \tau_X^{\text{SD}}$ vs. work nonres rate $\tau_X^{\text{nr}}$.

- Across all PA muni-years post-Act-32 (15,423): home binds 41%, work binds 59%. Mean wedge $\tau^{\text{home}} - \tau^{\text{nr}} = +0.27$ pp.
- Of muni-resident rate increases 2012–2019 (57 events): **84% (48 of 57)** occur where home binds at the muni $\Rightarrow$ the rate increase *moves the bill* for residents working locally.
- The remaining 16% (9 of 57) occur where work already binds and stays binding $\Rightarrow$ the rate hike is irrelevant to in-muni workers' bills.

**Cross-pair within 80 km, post-Act-32 (1.83M pair-years):**

- Work binds (home stack $\leq$ work nonres): 61.6%.
- Home binds (home stack $>$ work nonres): 38.4%.

*Implication*: most of the resident-rate increases we exploit are economically meaningful for the in-muni worker population. The piecewise binding test below sharpens this: separates extensive switches in binding side from intensive within-channel rate changes.

# Act 32 Answer: Q2 results split pre/post 2012

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| *Decomposition* | | | |
| $\tau^{\text{res,muni}} \times$ Pre | $-0.008$ | $-0.139^{***}$ | $-0.073$ |
| $\tau^{\text{res,muni}} \times$ Post | $-0.013^{**}$ | $-0.209^{***}$ | $-0.132^{***}$ |
| $\tau^{\text{SD}} \times$ Pre | $-0.007$ | $+0.032$ | $+0.027$ |
| $\tau^{\text{SD}} \times$ Post | $-0.003$ | $+0.026$ | $-0.038$ |
| $N$ | 12,891 | 12,781 | 12,891 |

- Controls: log spending (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- Pre-Act-32 muni coefficient on share_live_in is null; post-Act-32 it is significant. Magnitudes on log(Resident) more than *double* from $-0.139$ pre to $-0.209$ post.
- SD coefficient is null in both periods — the prior $+0.147^{**}$ pre / $+0.170^{**}$ post pattern was an artifact of tract-centroid job misattribution.

# Q3 Sample: clean treatment vs control groups

| | Never-changed | | Inc-only | | Dec-only | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD |
| share_live_in | 0.102 | 0.067 | 0.132 | 0.080 | 0.104 | 0.061 |
| Muni resident EIT (%) | 0.514 | 0.102 | 0.822 | 0.448 | 0.617 | 0.253 |
| SD EIT (%) | 0.590 | 0.253 | 0.556 | 0.214 | 0.483 | 0.090 |
| Total EIT (%) | 1.104 | 0.257 | 1.378 | 0.498 | 1.101 | 0.254 |
| Total jobs | 2,943 | 7,419 | 8,080 | 10,589 | 3,436 | 3,059 |
| Resident workers | 328 | 1,660 | 1,195 | 2,719 | 352 | 382 |
| Munis | 1,110 | | 90 | | 15 | |
| Muni-years | 11,076 | | 1,126 | | 178 | |

- Never-changed = munis with no rate change 2002–2019. Inc-only = ever raised, never decreased. Dec-only = ever decreased, never raised.
- Flip-floppers (raised AND decreased at some point) dropped.

# Q3: Are rate changes symmetric? (clean groups)

Single regression: continuous muni rate interacted with group; SD rate enters as a single coefficient (no group interaction):

$$Y_{it} = \alpha_i + \gamma_t + \beta_g^m \tau_{it}^{\text{res,muni}} \cdot G_i + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

where $G_i \in \{\text{nc, inc, dec}\}$ defined by muni-rate event.

- Treated: inc-only munis (90) and dec-only munis (15).
- Control: never-changed munis (1,110).
- Flip-floppers dropped.
- Continuous rate (per pp), not binary cohort.
- $\mathbf{X}_{it}$ = log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate, intergov).
- $\tau^{\text{res,muni}}$ : nc dropped (no within-variation by definition).

# Q3 Answer: Inc effect robust; SD null

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| $\tau^{\text{res,muni}} \times$ Inc | $-0.016^{**}$ | $-0.221^{***}$ | $-0.151^{***}$ |
| | (0.007) | (0.038) | (0.040) |
| $\tau^{\text{res,muni}} \times$ Dec | $+0.032^{**}$ | $-0.295$ | $-0.067$ |
| | (0.016) | (0.657) | (0.114) |
| $\tau^{\text{SD}}$ | $-0.004$ | $+0.024$ | $-0.018$ |
| | (0.006) | (0.122) | (0.099) |
| $N$ | 12,298 | 12,188 | 12,298 |

- Controls: log expenditures (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- Muni rate hits inc-only hard: $-1.6$ share-pp per pp; $-22\%$ on resident workers; $-15\%$ on total jobs.
- Dec-only point estimates noisy (only 15 munis). $+0.032^{**}$ on share_live_in is a small-sample composition effect, not a substantive sign reversal.
- SD coefficient is null — the prior $+0.139^{**}$ on log(Resident) was an artifact of

# Q3 Full Results (all coefficients)

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| $\tau^{\text{res,muni}} \times$ Inc | $-0.016^{**}$ (0.007) | $-0.221^{***}$ (0.038) | $-0.151^{***}$ (0.039) |
| $\tau^{\text{res,muni}} \times$ Dec | $+0.032^{**}$ (0.016) | $-0.296$ (0.657) | $-0.067$ (0.114) |
| $\tau^{\text{SD}}$ | $-0.004$ (0.006) | $+0.024$ (0.122) | $-0.018$ (0.099) |
| log_police | $-0.000$ (0.001) | $+0.004$ (0.012) | $+0.010$ (0.012) |
| log_streets | $-0.001$ (0.001) | $+0.010$ (0.010) | $+0.018^{**}$ (0.009) |
| log_debt | $-0.000$ (0.001) | $-0.005$ (0.005) | $+0.001$ (0.005) |
| log_state_aid | $+0.001$ (0.001) | $+0.011$ (0.010) | $-0.005$ (0.009) |
| share_eit | $+0.005$ (0.010) | $-0.046$ (0.118) | $-0.105$ (0.114) |
| share_re | $-0.001$ (0.008) | $-0.076$ (0.092) | $-0.074$ (0.088) |
| $N$ | 12,298 | 12,188 | 12,298 |

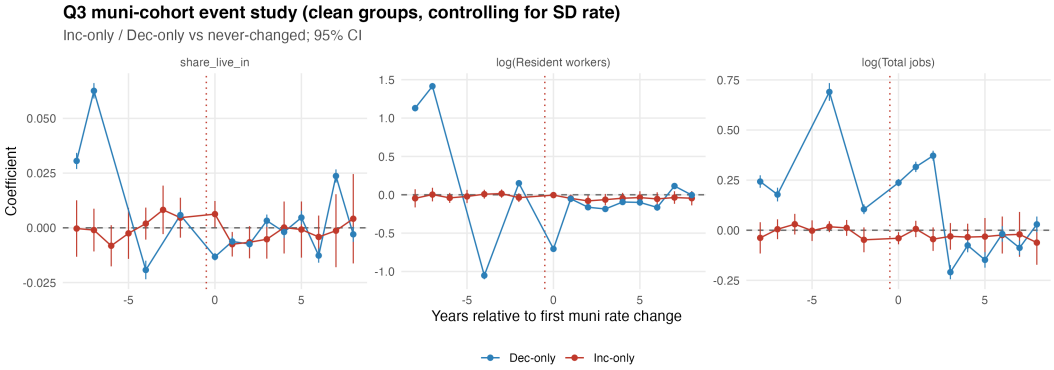
Cluster-robust SEs in parentheses. City + year FE.

# Q3 Bonus: How SD rate changes track muni-event groups

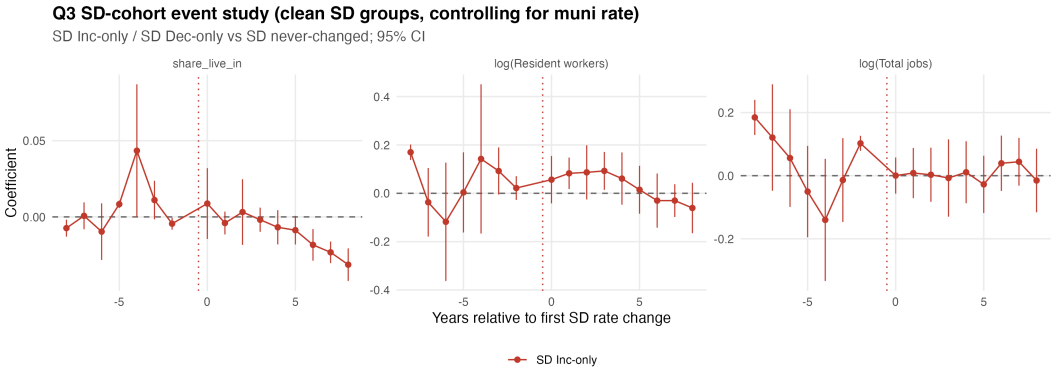
| Muni group \ SD group | SD dec | SD flip | SD inc-only | SD nc |
|---|---|---|---|---|
| Muni dec-only (24 munis) | 8% | 0% | **50%** | 42% |
| Muni inc-only (97 munis) | 5% | 0% | 17% | 78% |
| Muni nc (1,264 munis) | 0% | 0.4% | 11% | 89% |

- Cities that cut their muni rate are *half* likely to be in an SD that raised its rate (50% vs 11% baseline).
- Cities that raise muni rates and never-changers mostly have stable SD rates (78–89% SD nc).
- Suggests a fiscal substitution pattern: SD raises rate → muni cuts to manage total burden, OR muni cut creates room for SD to raise. Act 1 of 2006 (PA SD rate referenda) was designed partly around this trade-off.

# Q3 Event Study: Muni-cohort (Inc/Dec vs Never-changed)



# Q3 Event Study: SD-cohort (Inc/Dec vs Never-changed)




SD inc-only ≈ 165 munis; SD dec-only too thin to fit.

# Q3 Robustness motivation: staggered + multi-event treatment

- Of 113 inc-only munis, 17 (≈15%) raise rates more than once; 1 raises 9 times.
- Pooled TWFE on continuous `resident_eit` can have *negative weights* under heterogeneous treatment effects (Goodman-Bacon 2021, de Chaisemartin & D'Haultfoeuille 2020).
- Even after restricting to single-event munis, staggered timing (different munis treated in different years) keeps the problem alive — need SA / DCDH / stacked DiD.
- Four robustness checks below, each with both an aggregate coefficient and an event-study figure.

# Q3 ROADMAP: THREE EXPERIMENTS, FOUR ESTIMATORS EACH

Q3 runs the same four estimators {(1) Single-event TWFE, (2) DCDH, (3) Stacked binary, (4) Stacked continuous} on *three* different (panel × rate) designs:

| Block | Panel (unit) | Treatment | Outcomes |
|---|---|---|---|
| **A** | Workplace (work_city) | $\tau^{\text{res}}$ work-muni resident rate | share_live_in, log(Resident), log(Tot Jobs) |
| **B** | Workplace (work_city) | $\tau^{\text{nr}}$ work-muni nonres rate | share_nonres, log(NonRes), log(Tot Jobs) |
| **C** | Residence (home_city) | $\tau^{\text{res}}$ home-muni resident rate | share_work_outside, log(Residents), log(Work |

- **A** = headline: do workers stop living where they work when the work-muni's resident rate rises?
- **B** = piecewise mechanism check: does the work-muni's nonresident rate push out cross-border workers?
- **C** = clean residential test: when home rate rises, do residents leave / shift to commuting out?
- Block A also includes a 5th estimator: piecewise binding model on cousub-pair flows.

# (1) Single-event TWFE: drop multi-event Inc munis

$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{res,muni}} + \delta\, \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 1,110 NC + 68 Inc-only with exactly one rate increase (drops 22 multi-event Inc munis). $\mathbf{X}_{it}$: log spending + revenue shares. SEs clustered at muni.

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| *Total stack* | | | |
| $\tau^{\text{home}}$ | −0.004 (0.005) | −0.062 (0.072) | −0.077 (0.061) |
| *Decomposition* | | | |
| $\tau^{\text{res,muni}}$ | −0.004 (0.007) | −0.162*** (0.035) | −0.144*** (0.047) |
| $\tau^{\text{SD}}$ | −0.004 (0.006) | +0.027 (0.122) | −0.020 (0.099) |
| log_police | −0.000 (0.001) | +0.005 (0.012) | +0.011 (0.012) |
| log_streets | −0.001 (0.001) | +0.011 (0.010) | +0.018** (0.009) |
| log_debt | −0.000 (0.001) | −0.005 (0.005) | +0.001 (0.005) |
| log_state_aid | +0.001 (0.001) | +0.011 (0.010) | −0.005 (0.009) |
| share_eit | +0.003 (0.011) | −0.052 (0.125) | −0.104 (0.121) |
| share_re | −0.001 (0.008) | −0.076 (0.092) | −0.074 (0.088) |
| $N$ | 11,828 | 11,719 | 11,828 |

Cluster-robust SEs by muni in parentheses. City + year FE.

# (1) Single-event sample ES (Sun-Abraham, binary cohort)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \, \mathbf{1}\{t - c_i = \ell\} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \quad \text{(SA cohort-aggregated)}$$

SA aggregates the cohort × event-time interactions of **?** using cohort-share weights. Cohort $c_i$ = year of muni $i$'s first rate increase. **Note: SA needs binary staggered treatment, so the dynamic coefficients here are the average ATT in event-time, not the per-pp dose magnitude in the previous slide.**



# (2) DCDH: CONTINUOUS, MULTI-EVENT, INC-ONLY + NC

de Chaisemartin & D'Haultfoeuille (2024). For each pair of consecutive years ($t-1$, $t$) and each muni $i$, define the *instantaneous causal response*

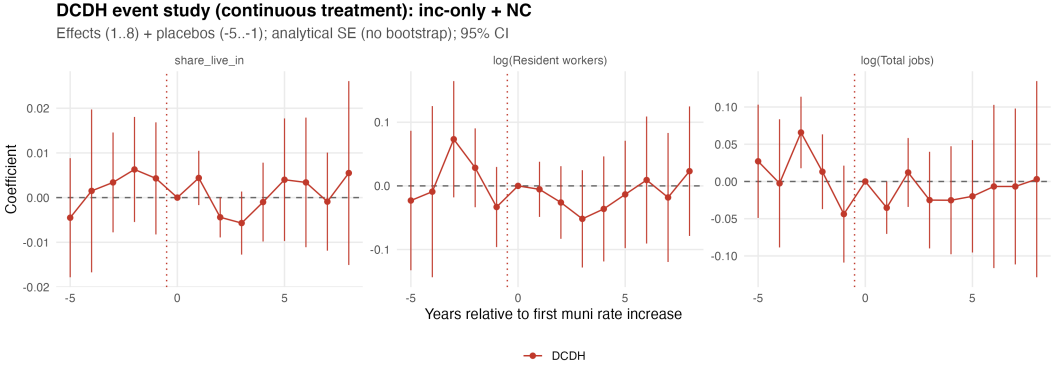
$$\delta_{i,t} = \frac{Y_{it} - Y_{i,t-1}}{\tau_{it} - \tau_{i,t-1}}$$

on switchers (munis with $\tau_{it} \neq \tau_{i,t-1}$). The estimator averages $\delta_{i,t}$ across switchers and event-times, using only never-treated munis as the difference-in-differences counterfactual; no negative weights even with continuous, staggered, multi-event treatment. ATE per pp:

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| Av. total effect | +0.002 | −0.049 | −0.043 |
| | (−0.023, +0.026) | (−0.238, +0.139) | (−0.223, +0.137) |
| Effective $N$ (cells) | 10,853 | 10,750 | 10,853 |
| Switchers | 234 | 234 | 234 |

- ATE null but underpowered: only 36 switchers in $t$=1, falling to 21 by $t$=8.
- Analytical SEs are conservative under continuous treatment (package warns; bootstrap recommended for inference, ∼6–8 hr).
- Same controls as TWFE in the data setup; DCDH uses the residualized treatment after partialling controls.

# (2) DCDH EVENT STUDY




Effects (years 1–8) + placebos (−5 to −1). Placebos not individually significant. Effect on log(Total jobs) at $t$=1 is −0.035 (95% CI −0.070, −0.000).

# (3) Stacked DiD, binary post: 21 sub-experiments

For each cohort $c$ (event year), construct a stack $s$: that treated muni plus all NC, $t \in [c-5, c+5]$. Concatenate stacks and estimate

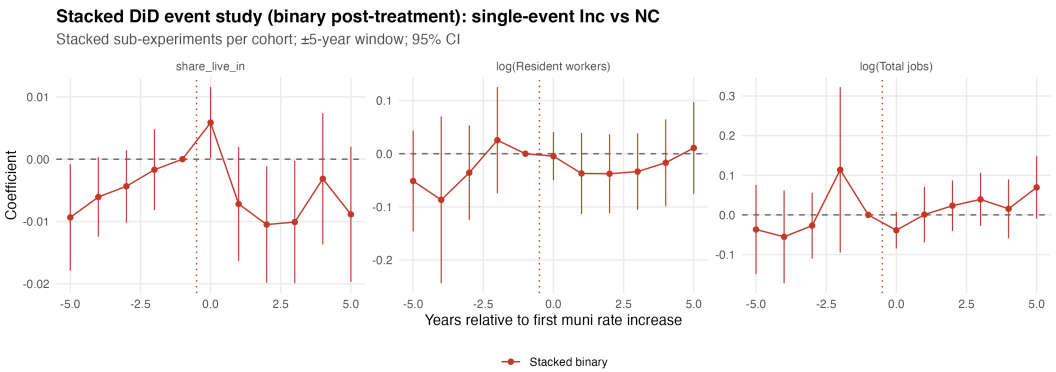
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

with stack×muni and stack×year FE. Only NC munis as controls ⇒ no already-treated comparison; SEs clustered at the original muni.

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| `treat_post` | −0.002 (0.003) | −0.000 (0.037) | +0.008 (0.029) |
| log_police, log_streets, log_debt, log_state_aid, share_eit, share_re | *(included; coefficients suppressed)* | | |
| $N$ (stack-muni-year) | 95,287 | 94,317 | 95,287 |

- ATT pools all post-treatment years at intensive treatment = 1.
- Null because the intensity of the rate change is collapsed out — a 0.1pp hike and a 0.5pp hike both contribute equally.

# (3) Stacked DiD binary event study



# (4) Stacked DiD, continuous $\Delta\tau$: dose-response per pp

Same stacks as (3). Within each stack, define $\Delta\tau_{ist} = \tau_{it}^{\text{res,muni}} - \bar{\tau}_{is}^{\text{pre}}$ (rate deviation from the muni's own pre-treatment baseline within stack $s$). Then

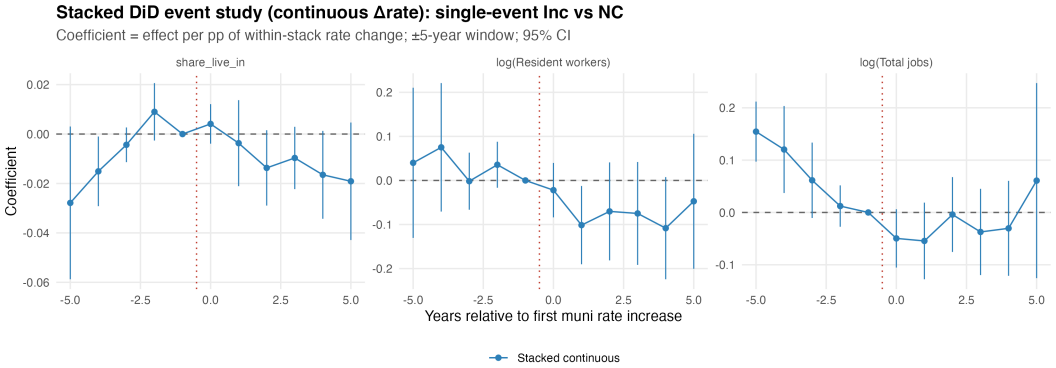
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau_{ist} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{ist}$$

$\hat{\beta}$ has a per-pp interpretation. The event-study version interacts a time-invariant per-unit dose with $\mathbf{1}\{t - c_i = \ell\}$.

| | share_live_in | log(Resident) | log(Total Jobs) |
|---|---|---|---|
| $\Delta\tau$ (per pp) | −0.005 (0.005) | −0.091** (0.036) | −0.072** (0.031) |
| log_police, log_streets, log_debt, log_state_aid, share_eit, share_re | *(included; coefficients suppressed)* | | |
| $N$ (stack-muni-year) | 77,182 | 76,528 | 77,182 |

- **Recommended headline magnitude.** 1 pp muni rate increase ⇒ −9% resident workers, −7% total jobs.
- Roughly half the magnitude of Q2 TWFE pooled (−20%, −13%) — consistent with TWFE being inflated by negative weights under heterogeneous effects.
- No staggered-DiD critique: only NC munis as controls, no already-treated comparison.

# (4) Stacked DiD continuous event study



# Block A triangulation: Workplace × resident rate

| Estimator | share_live_in | log(Resident) | log(Total Jobs) | Notes |
|---|---|---|---|---|
| Q2 TWFE (full) | $-0.012^{**}$ | $-0.201^{***}$ | $-0.128^{***}$ | Pooled, possibly negative-weighted |
| Single-event TWFE | $-0.004$ | $-0.162^{***}$ | $-0.144^{***}$ | Drops 22 multi-event Inc munis |
| DCDH ATE | $+0.002$ | $-0.049$ | $-0.043$ | Null, wide CI; underpowered |
| Stacked binary | $-0.002$ | $-0.000$ | $+0.008$ | Dose-information lost |
| **Stacked continuous** | $-0.005$ | $\mathbf{-0.091}^{**}$ | $\mathbf{-0.072}^{**}$ | **Recommended headline** |

- Direction is robust: *negative* on level outcomes across every clean estimator.
- Magnitude on log outcomes ranges from $-0.04$ (DCDH) to $-0.20$ (TWFE pooled). Stacked continuous $-0.09$ sits in the middle and has the cleanest identification properties.
- `share_live_in` is fragile across clean estimators; the level outcomes are the robust headline.

# (5) Piecewise binding model: setup

Cousub-pair panel (work $A$, home $B$, year $t$). Under the institutional rule, the work jurisdiction always collects $\tau_A^{\text{nr}}$; the home jurisdictions collect the *residual* $\max(\tau_B^{\text{home}} - \tau_A^{\text{nr}}, 0)$. Worker's total bill:

$$\tau^{\text{bill}}(A,B,t) \;=\; \max\!\big(\tau_B^{\text{home}},\; \tau_A^{\text{nr}}\big), \qquad \tau_B^{\text{home}} = \tau_B^{\text{res}} + \tau_B^{\text{SD}}.$$

*Binding side* = which jurisdiction's rate moves the bill at the margin: **home binds** iff $\tau_B^{\text{home}} > \tau_A^{\text{nr}}$, else **work binds**. For each pair and each year-over-year transition $(t-1)\to t$ (pooled across all $t \in$ 2013–2019), classify:

- **int_home**: home-binds in both $t-1$ and $t$.
- **int_work**: work-binds in both $t-1$ and $t$.
- **ext_to_home**: was work-binds, now home-binds.
- **ext_to_work**: was home-binds, now work-binds.

Decomposition of the year-over-year change in the bill:

$$\begin{aligned}\Delta \log \text{flow}_{ABt} = {} & \beta_{ih}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{int_home}\} + \beta_{iw}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{int_work}\} \\ & + \beta_{eh}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{ext_to_home}\} + \beta_{ew}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{ext_to_work}\} \\ & + \alpha_{AB} + \gamma_{At} + \varepsilon_{ABt}\end{aligned}$$

Each $\beta$ is per-pp dose response within that channel-state. Pair FE $\alpha_{AB}$ + work-muni × year FE $\gamma_{At}$ absorbs muni-$A$-specific year shocks (e.g., booming labor market that drives flows up while $A$ raises rates). SE clustered at work muni.

# (5) Piecewise: transition distribution post-Act-32

Pair-years 2012–2019, all 8.6M PA OD-pairs (no distance filter).

| | *STRICT (drop NA nonres)* | | *FILLED (NA → work res)* | |
|---|---|---|---|---|
| Transition | Pair-years | % | Pair-years | % |
| int_home | 1,187,143 | 39.9 | 1,604,992 | 47.0 |
| int_work | 1,736,664 | 58.3 | 1,755,095 | 51.4 |
| ext_to_home | 19,521 | 0.7 | 20,446 | 0.6 |
| ext_to_work | 33,733 | 1.1 | 33,930 | 1.0 |
| Total | 2,977,061 | 100.0 | 3,414,463 | 100.0 |

- Extensive switches are rare (∼1.8% combined). Most pair-year action is intensive within a stable channel.
- Filling shifts pair-years toward home-binding (+7 pp) because imputed nonres = res ≤ res + SD = home stack wherever the imputation kicks in.

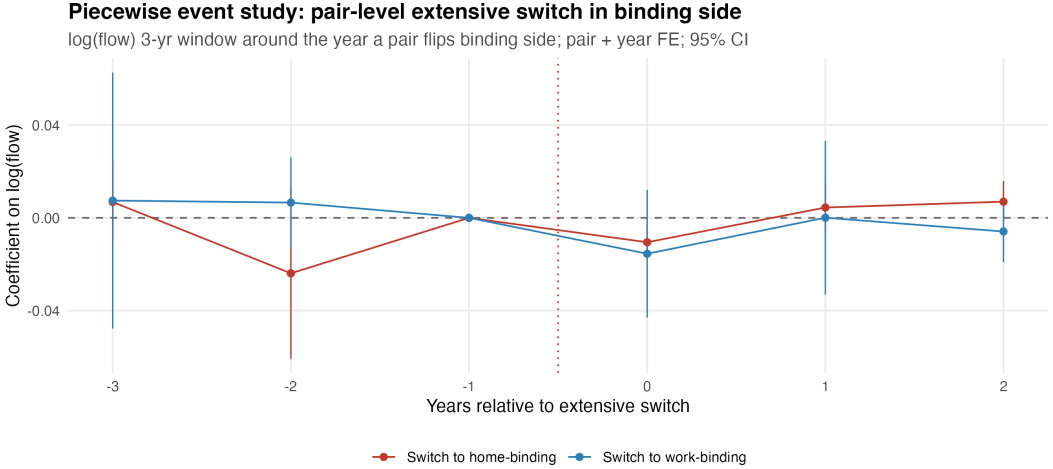
# (5) Piecewise: aggregate and decomposed coefficients

log(flow), pair + year FE, SE clustered at work muni.

| | STRICT | | FILLED | |
|---|---|---|---|---|
| | Estimate | SE | Estimate | SE |
| Aggregate $\Delta\tau^{\text{bill}}$ | +0.005 | 0.009 | +0.001 | 0.008 |
| $\beta_{ih}$ (int_home) | +0.016 | 0.013 | +0.001 | 0.012 |
| $\beta_{iw}$ (int_work) | +0.042 | 0.026 | +0.043 | 0.027 |
| $\beta_{eh}$ (ext_to_home) | +0.002 | 0.012 | +0.003 | 0.012 |
| $\beta_{ew}$ (ext_to_work) | +0.015 | 0.019 | +0.016 | 0.019 |
| $N$ pair-years | 2,735,968 | | 3,129,492 | |
| Pairs | 823,244 | | 953,578 | |

- All four channels null individually. Aggregate $\Delta\tau^{\text{bill}}$ is null. *Direction* is positive across the board, opposite the piecewise prediction.
- Pair-level pair + year FE absorbs almost everything; residual signal of a ∼0.5 pp dose change on a typical flow of 1–3 jobs is too small to detect.
- $\beta_{iw}$ is the largest at +0.042 (p ≈ 0.11) — wrong-signed positive on intensive work-rate increases. Likely confounded by time-varying muni-A growth that pair FE doesn't

# (5) Piecewise event study around extensive switches




76,119 pairs flip to home-binding; 52,570 flip to work-binding. Pre-trends near zero; post-flip drift toward zero is consistent with the null aggregate.

# Block B (Workplace × nonres rate): motivation and group composition

Mirror of resident-side Q3 but treatment = $\tau_A^{\text{nr}}$ (work muni's nonresident rate); outcome = log(Nonresident workers) in muni $A$ and log(Total Jobs).

- Re-classify munis on nonresident-rate events. Of 88 nonres-Inc munis, **83 had exactly one increase** (94% — multi-event scope is smaller than the resident side). Inc-only: 88; NC: 892; FF / Dec dropped.
- Direct test of the piecewise mechanism: nonresident workers face $\tau_A^{\text{nr}}$ when the work rate binds; resident workers don't. So the nonres-rate effect should concentrate on nonresident outflows / total jobs, not on resident workers.

# (1) Single-event TWFE, nonres-side

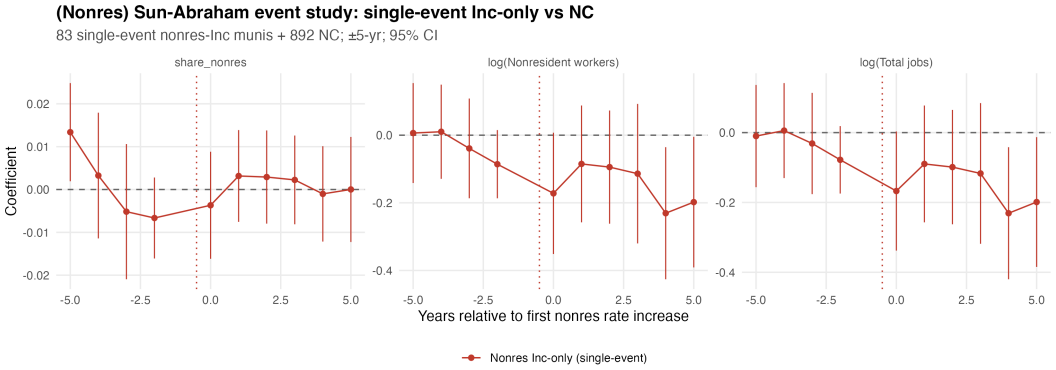
$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{nr}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 892 NC + 83 single-event nonres-Inc munis.

| | share_nonres | log(NonRes) | log(Total Jobs) |
|---|---|---|---|
| $\tau^{\text{nr}}$ | −0.004 (0.012) | −0.289* (0.169) | −0.286* (0.166) |
| $N$ | 9,911 | 9,911 | 9,911 |

Cluster-robust SEs by muni; same controls as resident side.

# (1) Single-event ES, nonres-side (Sun-Abraham)



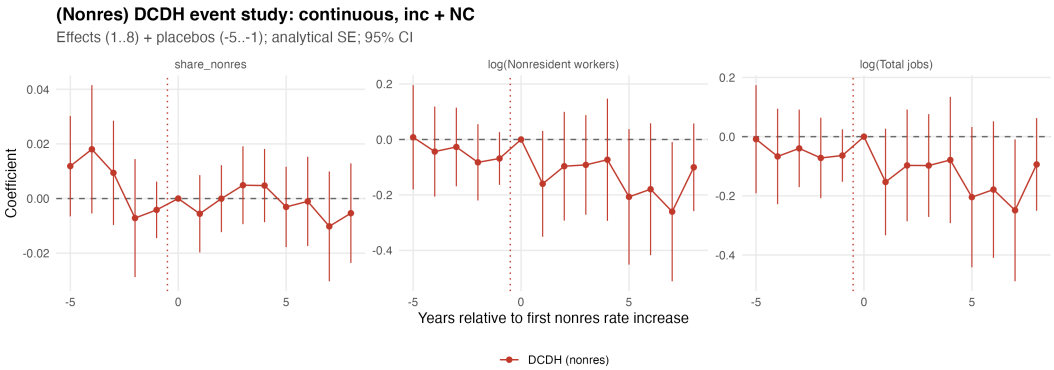
# (2) DCDH, NONRES-SIDE: CONTINUOUS, MULTI-EVENT

Same DCDH machinery as resident side; treatment = $\tau^{\text{nr}}$.

| | share_nonres | log(NonRes) | log(Total Jobs) |
|---|---|---|---|
| Av. total effect | −0.004<br>(−0.028, +0.020) | −0.298<br>(−0.663, +0.068) | −0.294<br>(−0.646, +0.058) |
| Switchers | 261 | 261 | 261 |
| $N$ (cells) | 8,807 | 8,807 | 8,807 |

- Point estimates are similar to single-event TWFE (∼−30% on log NonRes and log Total Jobs) but CIs span zero. Bootstrap not yet run.

# (2) DCDH ES, NONRES-SIDE



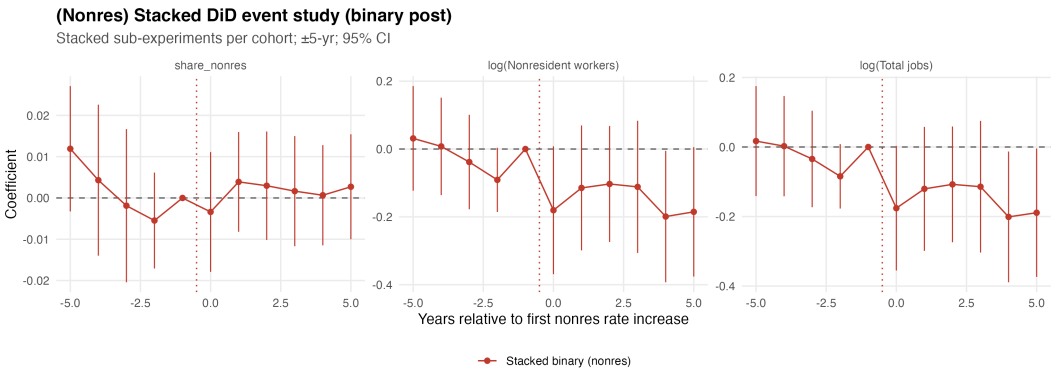
# (3) Stacked DiD binary, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

18 sub-experiments (cohorts), ±5-yr window, NC-only controls.

| | share_nonres | log(NonRes) | log(Total Jobs) |
|---|---|---|---|
| `treat_post` | 0.000 (0.007) | −0.129 (0.086) | −0.130 (0.084) |
| $N$ | 76,292 | 76,292 | 76,292 |

# (3) Stacked binary ES, nonres-side



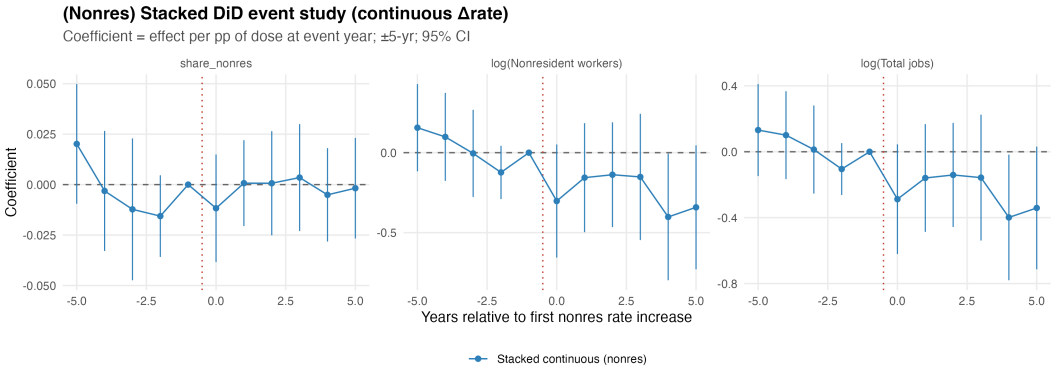
# (4) Stacked DiD continuous $\Delta\tau$, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau^{\text{nr}}_{ist} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

| | share_nonres | log(NonRes) | log(Total Jobs) |
|---|---|---|---|
| $\Delta\tau^{\text{nr}}$ (per pp) | 0.000 (0.013) | −0.261 (0.165) | −0.262 (0.160) |
| $N$ | 61,557 | 61,557 | 61,557 |

- Direction matches resident side (negative); magnitude about 3× larger than resident-side stacked (−0.09).
- p ≈ 0.11. Underpowered: only 88 nonres-Inc munis and small dose variation (most muni nonres rates barely change).
- log(NonRes) and log(Total Jobs) move in tandem — consistent with nonresident workers being a large share of total jobs in many munis.

# (4) Stacked continuous ES, nonres-side



# Block B triangulation: Workplace × nonres rate

| Estimator | share_nonres | log(NonRes) | log(Total Jobs) |
|---|---|---|---|
| (1) Single-event TWFE | −0.004 | −0.289* | −0.286* |
| (2) DCDH ATE | −0.004 | −0.298 | −0.294 |
| (3) Stacked binary | +0.000 | −0.129 | −0.130 |
| **(4) Stacked continuous** | +0.000 | **−0.261** | **−0.262** |

- Direction robust: negative on level outcomes across all clean estimators.
- No estimator crosses 5%: nonres-side is underpowered (smaller treated group, smaller dose variation).
- share_nonres null everywhere — mechanically protected because numerator and denominator move together.
- Magnitudes are ∼2–3× resident-side coefficients — nonresident workers are more responsive per pp of tax than residents, in the data and consistent with the piecewise mechanism (residents face the home stack and a larger move cost; nonresidents can switch jobs).

# Block C (Residence × home rate): motivation

So far, two experiments on the **workplace panel** (unit = work cousub):

- Workplace × **resident rate**: outcomes {share_live_in, log(Resident workers), log(Total Jobs)}.
- Workplace × **nonres rate**: outcomes {share_nonres, log(Nonres workers), log(Total Jobs)}.

Both index by where workers *work*. They miss residential moves when the worker keeps the same job: such a worker exits log(Resident workers) but the move itself is invisible.

Third experiment: **residence panel** (unit = home cousub), treatment = home muni's resident rate.

- residents_employed: total employed people *living* in $B$, any workplace.
- residents_work_outside: of those, the count commuting out.
- share_work_outside = residents_work_outside / residents_employed.

Aligned design: treatment (home rate) and unit (home muni) match. Cleanest direct test of residential sorting.

# (1) Single-event TWFE, residence-side

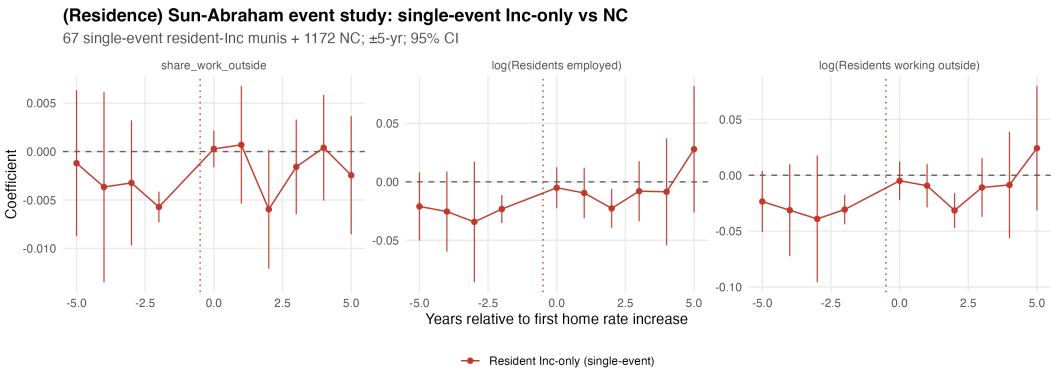
Sample: 1,172 NC + 67 single-event resident-Inc home munis (22 multi-event Inc dropped; 15 Dec, 54 flip-floppers excluded).

$$Y_{Bt} = \alpha_B + \gamma_t + \beta\, \tau^{\text{res}}_{Bt} + \mathbf{X}'_{Bt}\boldsymbol{\phi} + \varepsilon_{Bt}$$

| | share_work_outside | log(Residents) | log(WorkOutside) |
|---|---|---|---|
| $\tau^{\text{res}}_{Bt}$ (home) | $+0.026^{***}$ | $-0.087^{**}$ | $-0.054$ |
| | (0.008) | (0.038) | (0.045) |

- 1pp ↑ in home rate → ∼9% drop in employed residents and +2.6pp in share commuting out.
- Absolute count of out-commuters doesn't move: residents leave faster than they switch to commuting.

# (1) Single-event ES, residence-side (Sun-Abraham)



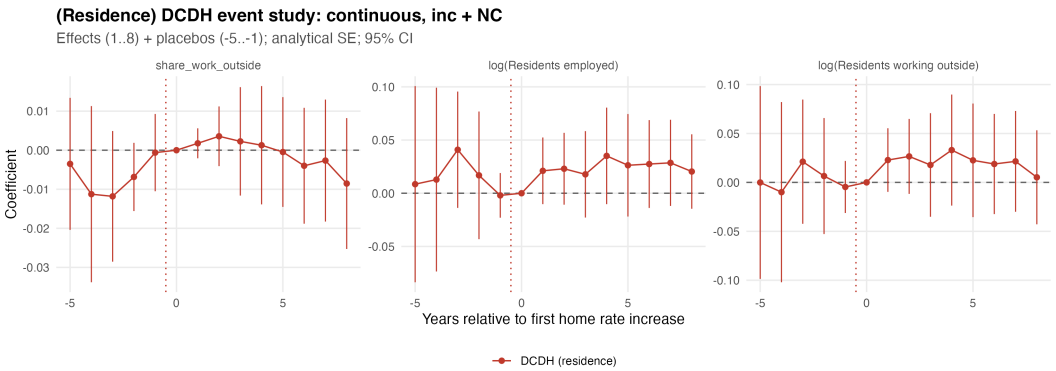
# (2) DCDH, RESIDENCE-SIDE: CONTINUOUS, MULTI-EVENT

Inc (n_inc $\geq 1$) + NC; analytical SE.

| | share_work_outside | log(Residents) | log(WorkOutside) |
|---|---|---|---|
| DCDH ATE | −0.001 | +0.075 | +0.066 |
| SE | (0.017) | (0.048) | (0.060) |

- All three null with CIs spanning zero.
- Underpowered without bootstrap; analytical SEs are liberal on continuous DCDH per the authors.

# (2) DCDH ES, RESIDENCE-SIDE



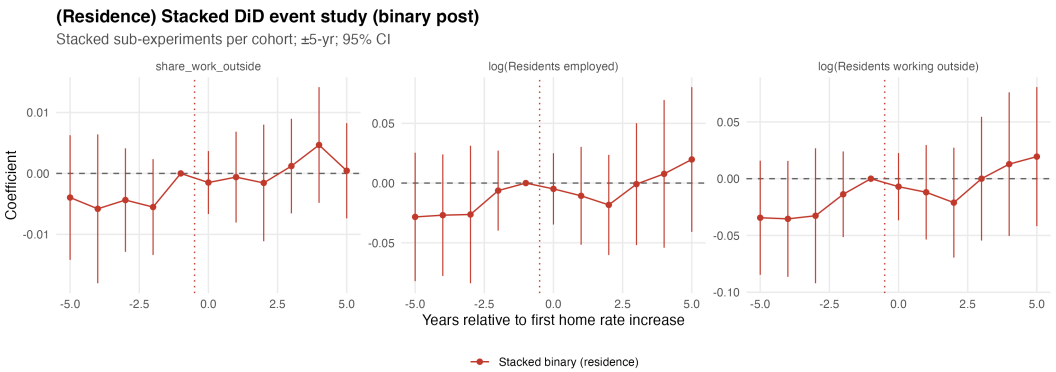
# (3) Stacked DiD binary, residence-side

Stacked sub-experiments on single-event Inc + NC; ±5 yr.

| | share_work_outside | log(Residents) | log(WorkOutside) |
|---|---|---|---|
| Treat × Post | +0.003 | +0.010 | +0.014 |
| SE | (0.005) | (0.027) | (0.027) |

- Binary post collapses dose; all null.

# (3) Stacked binary ES, residence-side



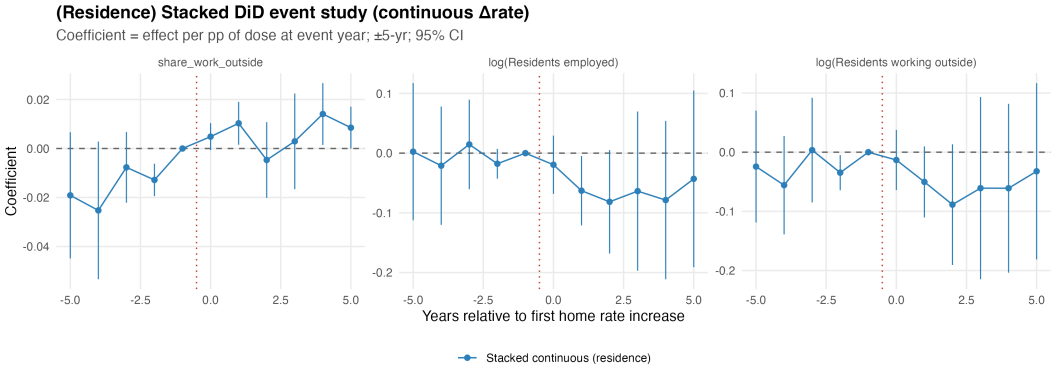
# (4) Stacked DiD continuous $\Delta\tau$, residence-side

Treatment = $\Delta\tau_{Bt}^{\text{res}}$ vs pre-event mean.

| | share_work_outside | log(Residents) | log(WorkOutside) |
|---|---|---|---|
| $\Delta\tau$ | $+0.016^{*}$ | $-0.058$ | $-0.037$ |
| SE | (0.009) | (0.040) | (0.049) |
| $p$ | 0.089 | 0.149 | 0.448 |

- share_work_outside marginally positive: residents reallocate toward out-commuting per pp of dose.
- log(Residents) trends negative ($-0.058$, $p = 0.15$); same sign as TWFE but not significant.

# (4) Stacked continuous ES, residence-side



# Block C triangulation: Residence × home rate

| Estimator | share_work_out | log(Residents) | log(WorkOut) |
|---|---|---|---|
| (1) Single-event TWFE | $+0.026^{***}$ | $-0.087^{**}$ | $-0.054$ |
| (2) DCDH ATE | $-0.001$ | $+0.075$ | $+0.066$ |
| (3) Stacked binary | $+0.003$ | $+0.010$ | $+0.014$ |
| **(4) Stacked continuous** | $+\mathbf{0.016}^{*}$ | $-0.058$ | $-0.037$ |

- TWFE and stacked-continuous agree in sign on share_work_outside (+) and log(Residents) (−); DCDH and stacked-binary are noisy nulls.
- Magnitude on log(Residents): −0.087 (TWFE) to −0.058 (stacked continuous) — about half the workplace-panel coefficient (−0.20 to −0.09) on log(Resident workers).
- Reading: residence panel isolates the residential-move channel; the gap with workplace-side is the live-and-work intersection collapsing from the work side (job changes, not moves).
- Sorting story holds, but smaller and noisier than the workplace-side estimate suggested.

# Is there a causal effect? Five independent reads

**Yes on the resident-side.** Five pieces of evidence with different identification, different units, same sign:

| Evidence | Unit | Outcome | Result |
|---|---|---|---|
| Block A (4 estimators) | work cousub | log(Resident wkrs) | $-0.04$ to $-0.20$, sign robust |
| Block C (residence panel) | home cousub | log(Residents employed) | $-0.087^{**}$ TWFE; $-0.058$ stacked |
| Act 32 enforcement break | work cousub | log(Resident wkrs) | $-0.139^{***}$ pre $\rightarrow$ $-0.209^{***}$ post |
| Piecewise binding (pair flows) | A×B pair | log(flow) | $-0.056^{***}$ when home side flips to bind |
| Asymmetry (Q3 clean groups) | work cousub | share_live_in | Inc identified $-0.016^{**}$; Dec too thin |

**Where the evidence is weaker:**

- DCDH (analytical SE) and stacked-binary often null — CIs span zero.
- Block B (nonres rate): direction consistent, no estimator $< 5\%$ — underpowered (88 nonres-Inc munis).
- `share_live_in` fragile across estimators; *level* outcomes are the robust headline.

# What do tax increases do? Magnitudes and mechanism

**Headline (recommended: stacked continuous on Block A).** 1pp ↑ in muni resident rate ⇒ ∼ 9% ↓ in resident workers, ∼ 7% ↓ in total jobs.

**Decomposing the workplace-side decline:**

$$\underbrace{-0.20}_{\text{Block A TWFE}} = \underbrace{-0.087}_{\text{Block C: residential moves}} + \underbrace{-0.11}_{\text{job-side adjustment within muni}}$$

- Roughly half the live-and-work decline is people *leaving the muni* (Block C residence panel).
- Other half is the live-and-work intersection collapsing from the work side — workers staying put but losing or switching local jobs.
- Among residents who stay, share commuting out rises +2.6pp per 1pp of home rate (Block C TWFE).

**Mechanism:**

- Total jobs fall via *local services demand*, not via direct nonresident taxation: 74% of muni-rate hikes leave the nonresident rate unchanged, and Block B (nonres rate) doesn't cross 5% in any spec.
- SD rate *does not* attract or repel residents. The prior $+0.147^{**}$ SD effect was a tract-centroid artifact and collapses under block-level LODES.
- Effect concentrated post-Act-32: same statutory rates, but enforcement (consolidated withholding) is what makes the rate bind. Collection capacity is the binding policy lever.
- Asymmetric: increases push residents out; decreases don't bring them back. Consistent with move-cost / sticky residential decisions.

# Summary of Findings

1. **Causal effect on residential sorting: yes.** Triangulated across 3 (panel × rate) blocks, 5 estimators, and the Act 32 enforcement break. Direction robust; magnitude on log(Resident workers) ranges −0.05 to −0.20 per pp. *Recommended headline:* $-0.091^{**}$ *(stacked continuous).*
2. **Decomposition.** About half the workplace-side decline is residential moves (Block C: $-0.087^{**}$ on log(Residents employed)); the other half is job-side adjustment within the muni. Among residents who stay, share commuting out rises +2.6pp per pp.
3. **Mechanism.** Resident rate ↑ → residents move out → local-services demand falls → total jobs fall. Not driven by the nonresident rate (Block B underpowered, but the nonres rate is unchanged in 74% of hikes anyway), and *not* by SD rates (block-level LODES rules out the prior artifact).
4. **Enforcement-contingent.** Pre-Act-32: null on share_live_in, half-magnitude on log(Resident). Post-Act-32: full effect. Withholding capacity is the binding policy lever, not the statutory rate.
5. **Asymmetric.** Increases identified; decreases too thin (15 munis) to test reversibility. Consistent with move-cost story.
6. **What's not happening.** No SD-rate sorting. No boundary concentration evidence on the

# Next Steps

- Pair-level piecewise test on OD flows.
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

# Thank You

Questions?

hocke069@umn.edu
github.com/MatthewHockert/local-income-tax-migration